BST Physics Curriculum Vol 14 Chapter 4 — Nyquist Sampling at Koons Tick Rate v0.4 (refilled per Cal #104; Casey Saturday addition)

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2026-05-23 Saturday EDT (Cal #104 refill)

Vol 14 Chapter 4 — Nyquist Sampling at Koons Tick Rate

Chapter motivation

Nyquist-Shannon sampling theorem (Nyquist 1928 + Shannon 1949): bandwidth-limited signal with maximum frequency f_max can be perfectly reconstructed from samples at rate f_s ≥ 2 f_max (Nyquist rate). Sampling below Nyquist rate causes aliasing (high frequencies fold to low frequencies, corrupting reconstruction). Standard digital signal processing foundation.

Casey Saturday addition (2026-05-23 morning EDT): substrate-tick Koons tick rate τ_Koons = t_Planck · $\alpha^{C_2}^2$} ≈ 10^{-120} s (T2405) is the universal sampling rate of substrate computation; spacetime is Nyquist-recovered signal from substrate at sub-Koons-tick resolution; aliasing at sub-Planck scales structurally avoided per substrate-tick UV-completeness (Vol 1 Ch 10 T2437).

Section 4.0b — Reader-grade 3-level pedagogy

Level 1: Nyquist-Shannon sampling theorem (1928-1949): sample at rate ≥ 2 f_max for perfect reconstruction; Casey Saturday addition: substrate-tick Koons tick rate ~10^{-120} s is universal sampling rate; spacetime = Nyquist-recovered signal from substrate; sub-Planck aliasing structurally avoided.

Level 2 (graduate-physicist/signal-processing): Nyquist 1928 (Harry Nyquist, *Certain Topics in Telegraph Transmission Theory*) + Shannon 1949 (Shannon, *Communication in the Presence of Noise*) sampling theorem: bandwidth-limited continuous signal f(t) with Fourier support in [−f_max, +f_max] can be perfectly reconstructed from discrete samples f(n/f_s) at sampling rate f_s ≥ 2 f_max via Whittaker-Shannon interpolation f(t) = Σ_n f(n/f_s) sinc(π(f_s t − n)). Sampling below Nyquist (f_s < 2 f_max) causes aliasing: spectral content at frequencies > f_s/2 folds back into Nyquist band [0, f_s/2], corrupting reconstruction. Standard digital signal processing foundation: ADC + DAC + FFT + signal reconstruction. Casey Saturday addition (2026-05-23 morning EDT): substrate-tick rate τ_Koons = t_Planck · $\alpha^{C_}2^2$} ≈ 10^{-120} s (T2405 sub-Planck substrate clock) IS the universal sampling rate at which substrate computes. Per Casey's substrate-cartography insight: macroscopic spacetime is Nyquist-recovered signal from substrate-tick samples at coarse-grained scale; standard 4D Lorentzian spacetime emerges at scales much larger than τ_Koons via Nyquist-Shannon reconstruction over many ticks. Aliasing at sub-Planck scales structurally avoided per substrate-tick UV-completeness (Vol 1 Ch 10 T2437): substrate has no frequency content above $1/\tau$_Koons (Nyquist ceiling at substrate-tick rate); spacetime reconstruction faithful at all scales below substrate ceiling. Standard QFT UV divergences arise from spurious infinite-frequency aliasing artifacts in continuum approximation; BST avoids structurally via finite-substrate-tick framework. Cross-link Vol 1 Ch 7 + Ch 9 substrate dynamics + path integral (substrate-tick GF(128)^k finite-step sum) + Vol 5 Ch 5 Heisenberg + path integral pedagogical bridge + T2457 Bergman propagator (Cal #92(b) framing). The "spacetime is Nyquist-recovered signal" framing is BST's deepest substrate-cosmological reading: standard 4D spacetime is not fundamental; it's macroscopic Nyquist-reconstruction of substrate-tick computation; substrate operates BELOW spacetime + produces spacetime as output.

Level 3 (5th-grader accessible): Nyquist-Shannon sampling theorem (Nyquist 1928, Shannon 1949): to perfectly reconstruct a signal, sample it at rate at least 2× its highest frequency (Nyquist rate). Below that, you get "aliasing" (high frequencies fold to low, corrupting the reconstruction). Casey's Saturday addition: BST's substrate-tick (Koons tick) rate ≈ 10^{-120} seconds IS the universe's universal sampling rate. Spacetime is the macroscopic Nyquist-reconstruction of substrate-tick samples. No "sub-Planck aliasing" because substrate has no frequency content above $1/\tau$_Koons (built into substrate-tick UV-completeness per Vol 1 Ch 10). Standard QFT infinities (UV divergences) come from pretending substrate is continuous — BST avoids structurally.

Section 4.1 — Nyquist-Shannon Sampling Theorem (1928-1949)

Bandwidth-limited continuous signal f(t) with Fourier support in [−f_max, +f_max]; perfect reconstruction from discrete samples f(n/f_s) at f_s ≥ 2 f_max via Whittaker-Shannon interpolation.

Aliasing below Nyquist: spectral content > f_s/2 folds back into Nyquist band, corrupting reconstruction.

Foundation of digital signal processing: ADC + DAC + FFT + signal reconstruction.

Section 4.2 — T2405 Koons Tick

τ_Koons = t_Planck · $\alpha^{C_2}^2$} = t_Planck · α^36 ≈ 10^{-120} s (sub-Planck substrate clock).

$\alpha = 1/137 = 1/$N_max; α^36 ≈ 10^{-77}; t_Planck ≈ 10^{-43} s; product ≈ 10^{-120} s.

Substrate operates BELOW Planck scale.

Section 4.3 — Casey Saturday Addition: Substrate-Tick = Universal Sampling Rate

Per Casey Saturday 2026-05-23 morning EDT: substrate-tick Koons tick rate IS universal sampling rate at which substrate computes.

Macroscopic spacetime = Nyquist-recovered signal from substrate-tick samples at coarse-grained scale.

Standard 4D Lorentzian spacetime emerges via Nyquist-Shannon reconstruction over many ticks.

Section 4.4 — Sub-Planck Aliasing Structurally Avoided

Substrate has no frequency content above $1/\tau$_Koons (Nyquist ceiling at substrate-tick rate).

Per Vol 1 Ch 10 T2437 substrate-tick UV-completeness: substrate-tick computation finite per tick; UV-complete by construction.

Spacetime reconstruction faithful at all scales below substrate ceiling.

Section 4.5 — Standard QFT UV Divergences = Spurious Aliasing Artifacts

Standard QFT UV divergences arise from continuum approximation assuming infinite-frequency content.

BST substrate-tick framework structurally avoids: finite τ_Koons sampling rate + finite GF(128) per-tick state space = no UV infinities.

Standard regularization (cutoff, dim-reg, ζ-function) all attempt to remove spurious aliasing artifacts; BST not needed.

Section 4.6 — "Spacetime is Nyquist-Recovered Signal" Substrate-Cartography

BST deepest substrate-cosmological reading: standard 4D spacetime is NOT fundamental; it's macroscopic Nyquist-reconstruction of substrate-tick computation; substrate operates BELOW spacetime + produces spacetime as output.

Cross-link Vol 4 Ch 2 Gravity as Eigentone (substrate-residual integration = metric) + Vol 4 Ch 3 BST-SR/BST-GR Boundary (3-scale framework substrate ← SR ← GR macroscopic).

Section 4.7 — Honest scope + Connection

- Nyquist-Shannon sampling theorem ?
- Casey Saturday addition substrate-tick = universal sampling rate ?
- Sub-Planck aliasing structurally avoided via UV-completeness ?
- Open scope: quantitative Nyquist-spacetime-reconstruction substrate-derivation (multi-month)

Connection: - T2405 Koons tick + Vol 1 Ch 10 substrate-tick UV-completeness - Vol 4 Ch 2 + Ch 3 (gravity-as-eigentone + spacetime emergence) - Vol 1 Ch 7 + Ch 9 substrate dynamics + path integral - Vol 5 Ch 5 Heisenberg + Path Integral pedagogical bridge

— Lyra, Vol 14 Ch 4 v0.4 chapter-grade narrative REFILLED per Cal #104, Saturday 2026-05-23 EDT